

1. test_register_login_logout_registerfaild

This test tests the registration process by creating a new user with and verifies successful login and logout operations. The test also checks how the system handles errors during registration, such as mismatched passwords, ensuring that proper error messages are displayed.

2. test_booking_process

This script tests the entire flight booking process end-to-end. It begins with new user registration, followed by searching for flights with valid input data. The script selects a flight from the search results, fills out detailed passenger information, including address and emergency contact details, and submits the booking form. It also validates the payment process by submitting payment details, handling the loading screen during payment processing, and verifying the redirection to the success confirmation page. The test ensures that users can navigate back to the homepage seamlessly after completing the booking.

3. test_history_and_select_seat

This script focuses on verifying the booking history and seat selection features. It ensures that users can access their booking history after logging in and validates that the booking history page is displayed correctly. The script also tests the seat selection process for a booked flight, ensuring that users can select a seat, confirm their choice, and see the correct seat details on the confirmation page. This test ensures that users can manage their bookings efficiently and that the application provides accurate feedback for seat selection and confirmation processes.

Test result:

```
PS C:\Projects\Group-6-FS> & C:/Users/New/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Projects/Group-6-FS/Flight_booking/selenium_test.py

DevTools listening on ws://127.0.0.1:10885/devtools/browser/fd41df9b-46d9-4fba-bee1-793cc3b11ecc
Starting Flask app...
127.0.0.1 - - [03/Dec/2024 15:03:37] "GET /register HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:37] "GET /static/register/images/background.jpg HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:37] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [03/Dec/2024 15:03:37] "POST /register HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:03:37] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:38] "GET /static/homepage/css/styles.css HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:38] "GET /static/homepage/images/Welcome%20to%20G6%20flights.png HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:38] "GET /static/homepage/images/Search.png HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:38] "GET /static/homepage/images/Background.jpg HTTP/1.1" 200 -
Registration successful.
127.0.0.1 - - [03/Dec/2024 15:03:39] "GET /logout HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:03:39] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:39] "GET /static/homepage/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:39] "GET /static/homepage/images/Welcome%20to%20G6%20flights.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:39] "GET /static/homepage/images/Search.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:39] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:40] "GET /register HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:40] "GET /static/register/images/background.jpg HTTP/1.1" 304 -
Password mismatch test passed.
127.0.0.1 - - [03/Dec/2024 15:03:42] "GET /login_page HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:42] "GET /static/login/globals.css HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:42] "GET /static/login/style.css HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:42] "GET /static/login/styleguide.css HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:42] "GET /static/login/images/background.jpg HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:43] "POST /login HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:03:43] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:43] "GET /static/homepage/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:43] "GET /static/homepage/images/Welcome%20to%20G6%20flights.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:43] "GET /static/homepage/images/Search.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:43] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
Login successful.
127.0.0.1 - - [03/Dec/2024 15:03:44] "GET /logout HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:03:44] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:44] "GET /static/homepage/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:44] "GET /static/homepage/images/Welcome%20to%20G6%20flights.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:44] "GET /static/homepage/images/Search.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:44] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
Logout successful.

Registration, registration password mismatch, login, and logout test passed.
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127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /register HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/register/images/background.jpg HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "POST /register HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/images/Welcome%20to%20G6%20flights.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/images/Search.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/images/Welcome%20to%20G6%20flights.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/images/Search.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:46] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:47] "POST /search-results HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:47] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:50] "POST /flight_detail HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:50] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
Flight selected.
127.0.0.1 - - [03/Dec/2024 15:03:52] "GET /passenger_info HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:52] "GET /static/register/images/background.jpg HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:03:54] "POST /passenger_info HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:55] "POST /process_payment HTTP/1.1" 200 -
Payment form submitted.
Loading process started.
127.0.0.1 - - [03/Dec/2024 15:03:59] "GET /finalize_payment HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:03:59] "GET /paymentsuccess HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:03:59] "GET /static/paymentsuccess/check.png HTTP/1.1" 200 -
Payment completed successfully.
127.0.0.1 - - [03/Dec/2024 15:04:00] "GET /home HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:04:00] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:04:00] "GET /static/homepage/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:04:00] "GET /static/homepage/images/Welcome%20to%20G6%20flights.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:04:00] "GET /static/homepage/images/Search.png HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:04:00] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
Clicked 'back to homepage' button.
Successfully navigated to the home page.

Booking process test passed

127.0.0.1 - - [03/Dec/2024 15:04:02] "GET /booking-history HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:04:02] "GET /static/bookinghistory/css/booking_history.css HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:04:02] "GET /static/homepage/images/Background.jpg HTTP/1.1" 304 -
Booking history page loaded successfully.
127.0.0.1 - - [03/Dec/2024 15:04:05] "GET /select_seat/2e7c44d2-8643-46c7-98ac-ca5a6502eb3c HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:04:05] "GET /static/selectseatsstyle.css HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:04:05] "GET /static/register/images/background.jpg HTTP/1.1" 304 -
Seat selection page loaded successfully.
Seat 5B selected successfully.
127.0.0.1 - - [03/Dec/2024 15:04:08] "POST /confirm_seat/2e7c44d2-8643-46c7-98ac-ca5a6502eb3c HTTP/1.1" 302 -
127.0.0.1 - - [03/Dec/2024 15:04:08] "GET /seat_confirmation/2e7c44d2-8643-46c7-98ac-ca5a6502eb3c/5B HTTP/1.1" 200 -
127.0.0.1 - - [03/Dec/2024 15:04:08] "GET /static/selectseatsstyle.css HTTP/1.1" 304 -
127.0.0.1 - - [03/Dec/2024 15:04:08] "GET /static/register/images/background.jpg HTTP/1.1" 304 -
Seat 5B confirmed successfully on confirmation page.

view booking history and select seat process test passed

```

How to run:

Run the following command to install the required libraries:

```
pip install selenium flask
```

Additionally, ensure you have the WebDriver (the WebDriver for your preferred browser) installed and compatible with your browser version.

and use *python end_to_end_test.py* to run the test.